



Sydney Girls High School

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Student Number

2023

YEAR 11

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided on pages 13 -15
- Write your student number on the multiple choice answer sheet and the cover page

Total marks – 35

Part A – Pages 2-5

10 marks

- Attempt Questions 1 – 10

Part B – Pages 6-8

13 marks

- Attempt Questions 11 – 12

Part B – Pages 9-11

12 marks

- Attempt Questions 13 – 15

Part A – 10 marks

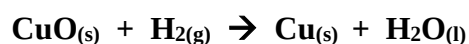
Attempt Questions 1 – 10

Use the multiple-choice answer sheet for Questions 1–10

- 1 Which observation will be made when a clean strip of magnesium is dipped into aqueous copper sulfate:

- (A) Hydrogen gas is released.
- (B) Sulfur dioxide gas is released.
- (C) The blue colour of the copper nitrate becomes more intense.
- (D) The magnesium strip becomes coated with a reddish deposit of copper.

- 2 Hydrogen and copper (II) oxide react to produce metal in water.



Which row correctly describes the oxidising and reducing agent?

	Oxidising agent	Reducing agent
(A)	H ₂ O	Cu
(B)	H ₂	CuO
(C)	Cu	H ₂ O
(D)	CuO	H ₂

- 3 Which combination of the following periodic table trends would result in an increase in a metal's reactivity?

- (A) Decrease in atomic radius, decrease in electronegativity.
- (B) Increase in ionisation energy, increase in atomic radius.
- (C) Increase in ionisation energy, decrease in electronegativity.
- (D) Decrease in ionisation energy, decrease in electronegativity.

4 Which equation represents a neutralisation reaction?

- (A) $2\text{H}_{2(g)} + \text{O}_{2(g)} \rightarrow 2\text{H}_2\text{O}_{(l)}$
- (B) $\text{C}_6\text{H}_{12}\text{O}_{6(aq)} + 6\text{O}_{2(g)} \rightarrow 6\text{CO}_{2(g)} + 6\text{H}_2\text{O}_{(g)}$
- (C) $\text{CuCl}_{2(aq)} + 2\text{NaOH}_{(aq)} \rightarrow \text{Cu}(\text{OH})_{2(s)} + 2\text{NaCl}_{(aq)}$
- (D) $2\text{HNO}_{3(l)} + \text{Ba}(\text{OH})_{2(aq)} \rightarrow \text{Ba}(\text{NO}_3)_{2(aq)} + 2\text{H}_2\text{O}_{(l)}$

5 Which of the following is a true statement that can be used to predict the entropy change in a system?

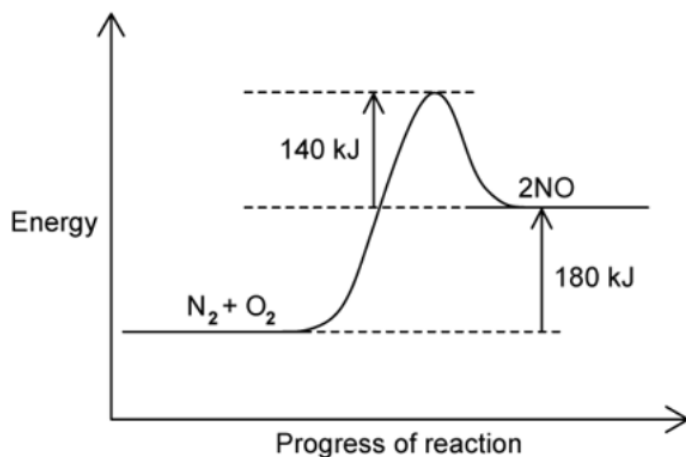
- (A) As the temperature of a substance increase, the entropy increases.
- (B) A substance in its gaseous state has less entropy, than the same amount of the substance in its liquid state.
- (C) A substance in its liquid state has less entropy than the same amount of substance in its solid state.
- (D) When a solute dissolves in a solvent, the solute particles in solution can be arranged in fewer ways than when they were solid, so the amount of entropy usually decreases

6 Catalysts such as iron and nickel are commonly used in industrial reactions.

Which statement about a catalyst is NOT correct?

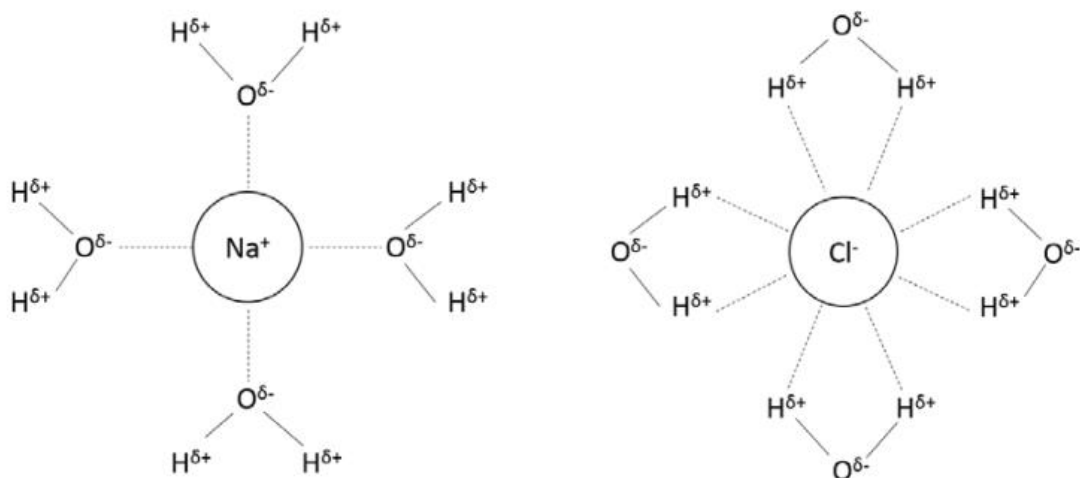
- (A) They create a pathway which has a lower activation energy.
- (B) Their concentration does not change throughout the reaction.
- (C) They increase the amount of energy required for a reaction to occur.
- (D) They can allow reaction to occur at much lower temperatures.

- 7 What is the activation energy for this reaction?



- (A) +140 kJ
- (B) +180 kJ
- (C) +320 kJ
- (D) -320 kJ
- 8 Which of the following is a statement describing Hess' Law?
- (A) Energy cannot be created nor destroyed, but it can be changed from one form into another.
- (B) In a chemical reaction, the total mass of the reactants is equal to the total mass of the products.
- (C) The enthalpy change for a chemical reaction is the same regardless of the number of steps involved in the process.
- (D) Enthalpy change in a chemical reaction equals the sum of the enthalpy of all products minus the sum of enthalpies of all reactants.

9 Diagram and models are often used to demonstrate complex chemical ideas.



Which chemical process is demonstrated by these diagrams?

- (A) Ionisation.
- (B) Carbonation.
- (C) Precipitation.
- (D) Dissociation.

10. Which row in the table shows how the metal will react?

	Metal	Reacts with dilute acid	Reacts rapidly with cold water	Reacts with steam
(A)	Copper	✓	✗	✗
(B)	Calcium	✗	✓	✓
(C)	Zinc	✓	✗	✗
(D)	Magnesium	✓	✗	✓

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Student Number

Part B – 25 marks

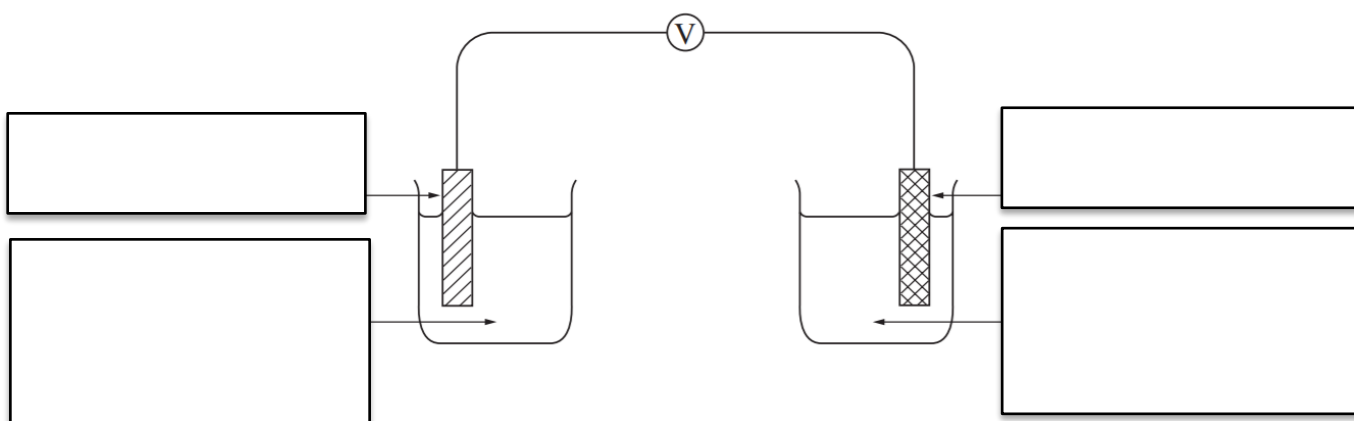
Attempt Questions 11 - 15

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 11 (5 marks)

The galvanic cell below is made by connecting two half-cells. One half-cell was made by putting an aluminium electrode in an aluminium nitrate solution. The other half-cell was made by putting a lead electrode in a lead nitrate solution. The electrodes were connected to a voltmeter as shown in the diagram.



Add the following labels to the diagram:

3P

- Which metal is the Anode and cathode
- Direction of electron flow
- Direction of anion and cation flow
- The two nitrate solutions
- A salt-bridge

- a) Using the standard potentials table in the data sheet, calculate the theoretical voltage of this galvanic cell.

2K

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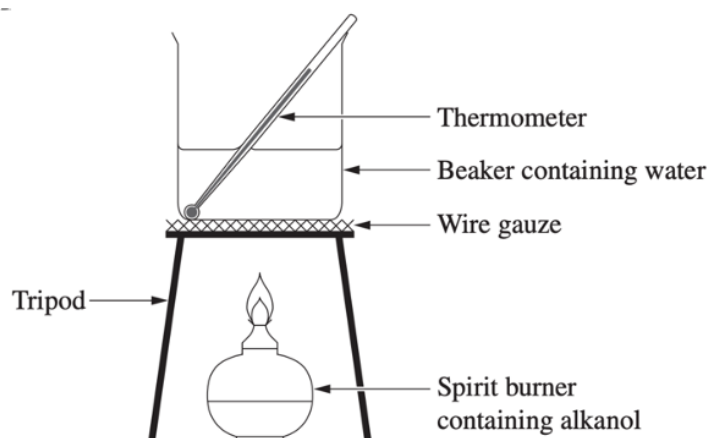
Question 12 (7 marks)

Three groups of students set out to determine the heats of combustion of the following alcohols:

- Methanol, CH_3OH
- Ethanol, $\text{C}_2\text{H}_5\text{OH}$
- Propan-1-ol, $\text{C}_3\text{H}_7\text{OH}$

Each group of students measured out 100mL of water into a container and heated the water by burning a measured mass of each of the alcohols.

Their results are shown in the table below:



Fuel (alcohol)	Amount of water heated (g)	Temperature rise (K)	Mass of alcohol burnt (g)	Heat of combustion ($\text{kJ}\cdot\text{mol}^{-1}$)
Methanol	100	10	0.185	725
Ethanol	100	10	0.142	?
Propan-1-ol	100	10	0.125	2016

- a) From the data in the table, calculate the molar heat of combustion of ethanol. 4K

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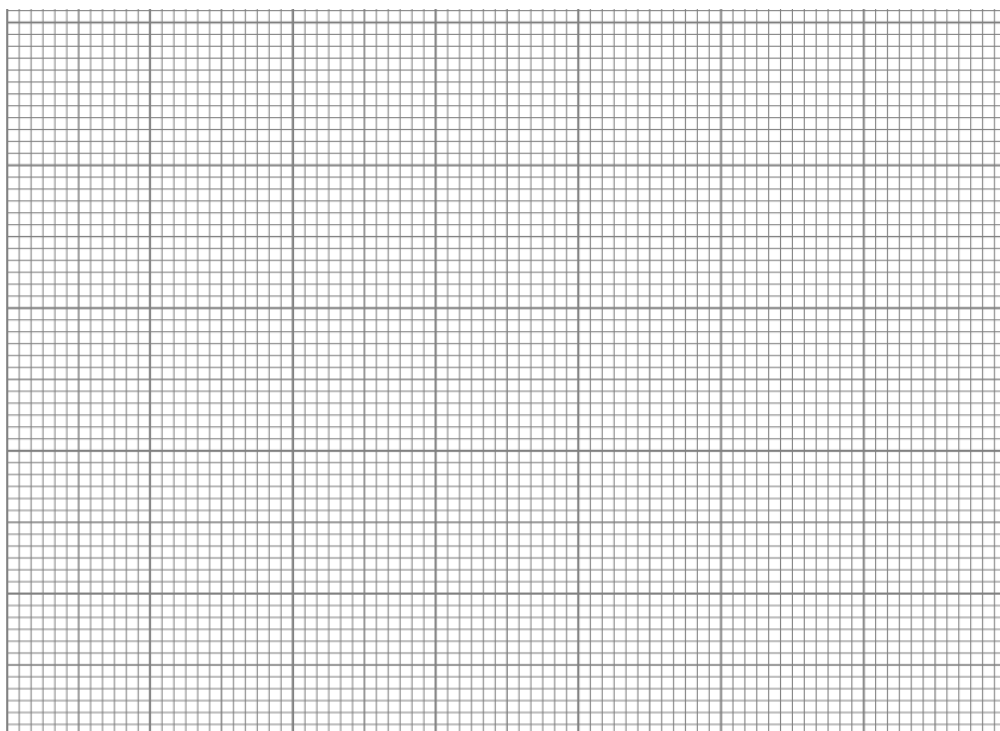
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- b) On the grid below, draw a graph representing the molar heat of combustion for the three alcohols against the molar mass of the alcohols. 3A



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Student Number

Part C – 13 marks

Attempt Questions 13 - 15

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 12 (5 marks)

- a) Write a method to describe an experiment you performed in a school laboratory to measure the change in reaction rate with concentration. 3P

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- b) Using collision theory, explain why increasing the temperature increases the rate of any chemical reaction. 2K

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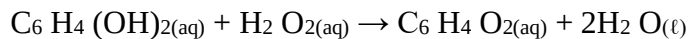
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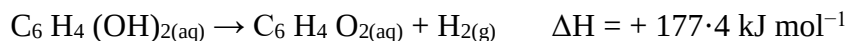
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Question 14 (4 marks)

The overall equation representing the formation of quinone, $\text{C}_6\text{H}_4\text{O}_2$ is as follows:



Use the following data to calculate the enthalpy change, in kJ mol^{-1} , for the above reaction. 4A



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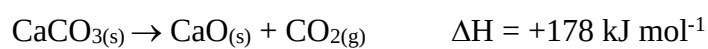
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Question 15 (4 marks)

Use the data in the table to answer this question.

	CaCO_{3(s)}	CaO_(s)	CO_{2(g)}
S (J mol⁻¹ K⁻¹)	90	40	214

Calcium carbonate decomposes according to the following equation at 25°C.



Calculate ΔG for this reaction.

4K

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Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_0}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and
at 0°C (273.15 K) 22.71 L
at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}(s)$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}(s)$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}(s)$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}(s)$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}(s)$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}(s)$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}(s)$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}(s)$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}(s)$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}(s)$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}(s)$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}(s)$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g)$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(aq) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.34 V
$\frac{1}{2}\text{O}_2(g) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.52 V
$\frac{1}{2}\text{I}_2(s) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(aq) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}(s)$	0.80 V
$\frac{1}{2}\text{Br}_2(l) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(g) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(g) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(g) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

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1 H Hydrogen 2 He Helium									
3 Li Lithium 4 Be Beryllium									
11 Na Sodium 12 Mg Magnesium									
19 K Potassium 20 Ca Calcium									
37 Rb Rubidium 38 Sr Strontium									
55 Cs Cesium 56 Ba Barium									
87 Fr Francium 88 Ra Radium									
9 F Fluorine 10 Ne Neon									
17 Cl Chlorine 18 Ar Argon									
25 Mn Manganese 26 Fe Iron									
33 As Arsenic 34 Se Selenium									
51 Sb Antimony 52 Te Tellurium									
83 Bi Bismuth 84 Po Polonium									
113 Nh Nihonium 114 Fl Flerovium									
115 Mc Moscovium 116 Lv Livermorium									
117 Ts Tennessine 118 Og Oganesson									
63 Eu Europium 64 Gd Gadolinium									
71 La Lanthanum 72 Ce Cerium									
89 Ac Actinium 90 Th Thorium									
91 Pa Protactinium 92 U Uranium									
93 Np Neptunium 94 Pu Plutonium									
95 Am Americium 96 Cm Curium									
97 Bk Berkelium 98 Cf Californium									
101 Md Mendelevium 102 No Nobelium									
103 Lr Lawrencium 104 Rf Rutherfordium									

Standard atomic weights are abridged to four significant figures.

Fillments with no reported values in the table have no stable nuclides.

Elements with an asterisk (*) indicate that the data were taken from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (November 2016 version). The information on elements with atomic numbers 113 and above is sourced from the International Union of Pure and Applied Chemistry Periodic Table of the Elements (February 2010 version) is the principal source of all other data. Some data may have been modified.



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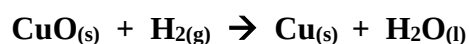
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- (C) $\text{CuCl}_{2(aq)} + 2\text{NaOH}_{(aq)} \rightarrow \text{Cu}(\text{OH})_{2(s)} + 2\text{NaCl}_{(aq)}$
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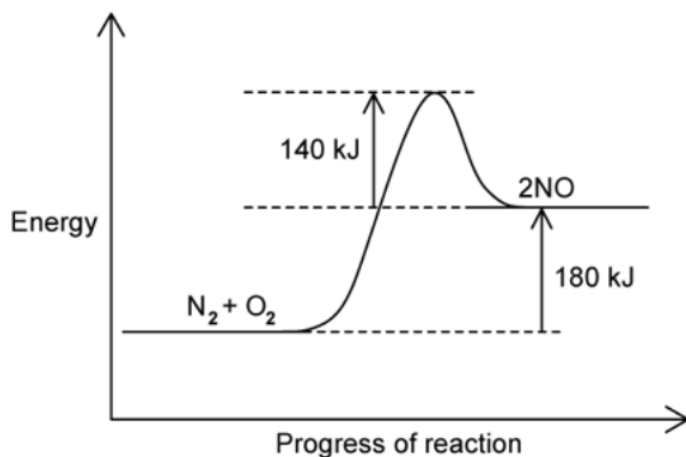
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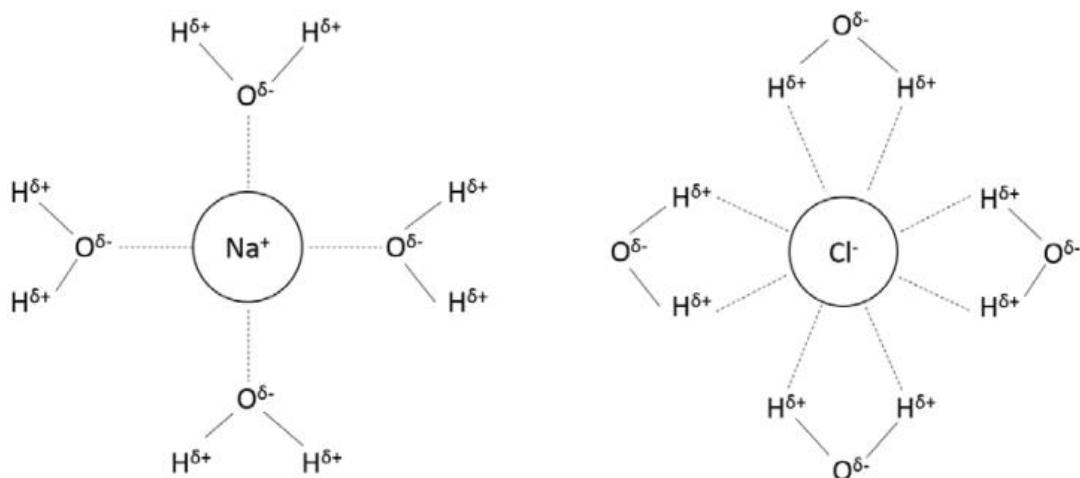


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9 Diagram and models are often used to demonstrate complex chemical ideas.



Which chemical process is demonstrated by these diagrams?

- (A) Ionisation.
- (B) Carbonation.
- (C) Precipitation.
- (D) **Dissociation.**

10. Which row in the table shows how the metal will react?

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(C)	Zinc	✓	✗	✗
(D)	Magnesium	✓	✗	✓

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Student Number

Part B – 25 marks

Attempt Questions 11 - 15

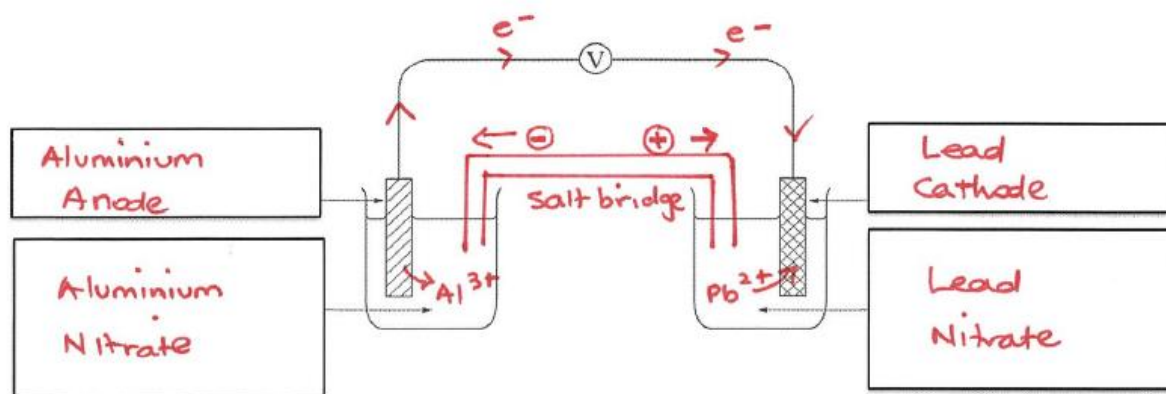
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Show all relevant working in questions involving calculations.

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Correct labels	3
Substantially correct labels	2
Some relevant labels	1



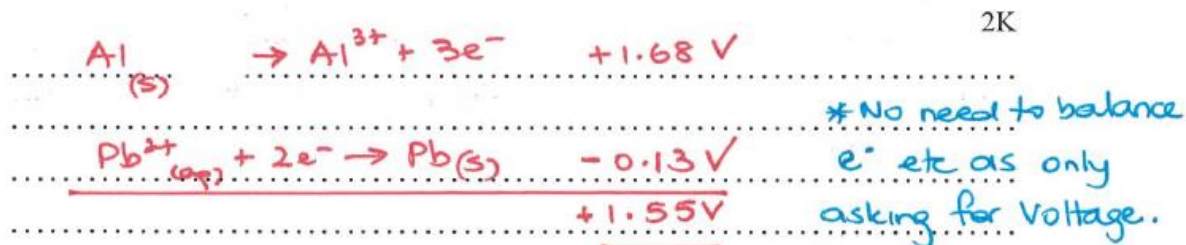
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- The two nitrate solutions
- A salt-bridge

- a) Using the standard potentials table in the data sheet, calculate the theoretical voltage of this galvanic cell. 2K

Correct Calculation	2
One relevant step in calculation	1



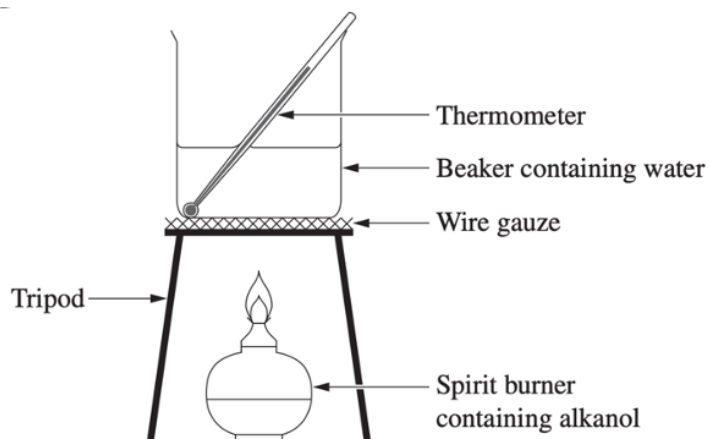
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a) From the data in the table, calculate the molar heat of combustion of ethanol. 4K

Any two parts of calculation performed eg • Calculation of heat supplied (eg q) • Correct calculation molar mass • Calculation of moles of ethanol • Calculation of ΔH from q (including – ve sign for exo rxn)	4
Most steps correct	3
Some relevant calculations	2
One relevant calculation	1

$$\begin{aligned}
 q &= mc\Delta T \\
 &= (100)(4.18)(10) \\
 \therefore q &= 4180 \text{ J or } 4.180 \text{ kJ} \\
 n &= m/M \\
 &= 0.142/46.068 \\
 \therefore n &= 0.0031 \text{ mol} \\
 \Delta H &= -q/n \\
 &= -4.180/0.0031 \\
 \therefore \Delta H &= -1348 \text{ kJ/mol}
 \end{aligned}$$

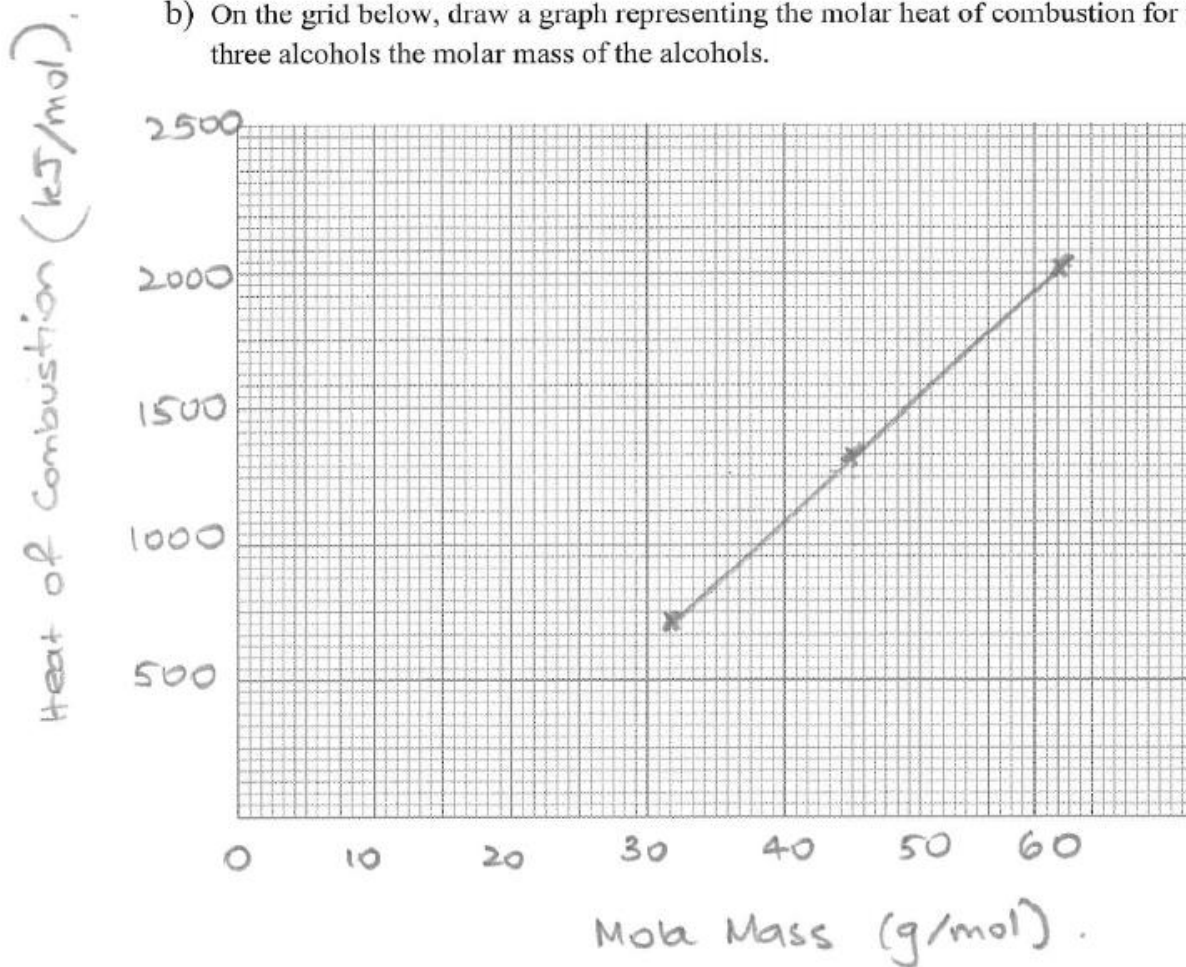
Graphed correctly including:

3

• Labels of both axes with correct units • Using > 50% of grid (appropriate linear scale used) • Correct plotting of points • Curve straight line	
As above but with 1 omission/error	2
As above but with 2 omissions/errors	1

b) On the grid below, draw a graph representing the molar heat of combustion for the three alcohols the molar mass of the alcohols.

3A



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Student Number

Part C – 13 marks

Attempt Questions 13 - 15

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 12 (5 marks)

- a) Write a method to describe an experiment you performed in a school laboratory to measure the change in reaction rate with concentration.

3P

Correct description of the method	3
Substantially correct description of the method	2
Some relevant steps	1

1. Measure + Place 50 mL of $\text{Na}_2\text{S}_2\text{O}_3$ into a 250 mL conical flask
2. Place the flask over a cross marked on white paper
3. Measure 5 mL of 1M HCl, add to the $\text{Na}_2\text{S}_2\text{O}_3$ in the conical flask; start the stopwatch
4. Stop the watch and record the time taken for the cross to become invisible when viewed from above
5. Record your results in a data table.
6. Repeat procedure using 40 mL, 30 mL, 20 mL, 10 mL of the $\text{Na}_2\text{S}_2\text{O}_3$ solution made up to a total volume of 50 mL with distilled water.

- b) Using collision theory, explain why increasing the temperature increases the rate of any chemical reaction.

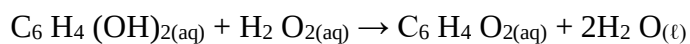
2K

Correct explanation	2
Partially correct explanation	1

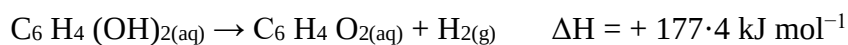
In collision theory, temp affects the rate of collision as well as the energy of the molecules. Higher temps $\uparrow$ the KE of the molecules so more molecules have the minimum KE (or $\geq$ than E_a). This increases the rate of the reaction.

Question 14 (4 marks)

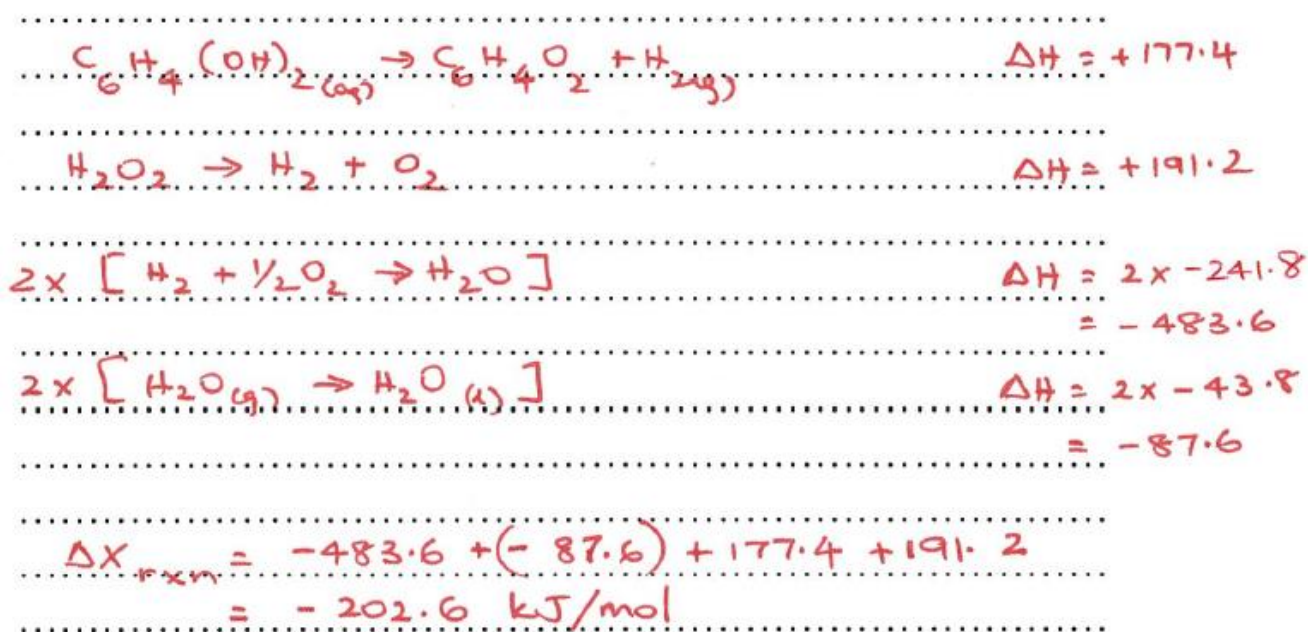
The overall equation representing the formation of quinone, $C_6H_4O_2$ is as follows:



Use the following data to calculate the enthalpy change, in kJ mol^{-1} , for the above reaction. 4A



All correct calculations, including significant figures	4
All steps correct	3
Several relevant calculations	2
Some relevant information	1



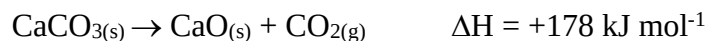
Question 15 (4 marks)

Use the data in the table to answer this question.

	$\text{CaCO}_{3(\text{s})}$	$\text{CaO}_{(\text{s})}$	$\text{CO}_{2(\text{g})}$
--	-----------------------------	---------------------------	---------------------------

S (J mol ⁻¹ K ⁻¹)	90	40	214
--	----	----	-----

Calcium carbonate decomposes according to the following equation at 25°C.



Calculate ΔG for this reaction.

4K

All correct calculations	4
Most steps correct	3
Some relevant calculations	2
One relevant calculation	1

$$\begin{aligned} \Delta S &= \text{products} - \text{reactants} \\ &= (40 + 214) - 90 \\ \therefore \Delta S &= +164 \text{ J mol}^{-1} \text{ K}^{-1} \\ &= +0.164 \text{ kJ mol}^{-1} \text{ K}^{-1} \\ \Delta H &= +178 \text{ kJ mol}^{-1} \\ T &= 298.15 \text{ K} \\ \Delta G &= +178 - (298.15 \times 0.164) \\ &= +129.103 \text{ kJ} \\ \therefore \Delta G &= +129 \text{ kJ} \quad (2 \text{ sig figs}). \end{aligned}$$

Chemistry

FORMULAE SHEET

$$n = \frac{m}{MM}$$

$$q = mc\Delta T$$

$$pK_a = -\log_{10}[K_a]$$

$$c = \frac{n}{V}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$A = \epsilon lc = \log_{10} \frac{I_0}{I}$$

$$PV = nRT$$

$$\text{pH} = -\log_{10}[\text{H}^+]$$

Avogadro constant, N_A $6.022 \times 10^{23} \text{ mol}^{-1}$

Volume of 1 mole ideal gas: at 100 kPa and
at 0°C (273.15 K) 22.71 L
at 25°C (298.15 K) 24.79 L

Gas constant $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Ionisation constant for water at 25°C (298.15 K), K_w 1.0×10^{-14}

Specific heat capacity of water $4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

DATA SHEET

Solubility constants at 25°C

Compound	K_{sp}	Compound	K_{sp}
Barium carbonate	2.58×10^{-9}	Lead(II) bromide	6.60×10^{-6}
Barium hydroxide	2.55×10^{-4}	Lead(II) chloride	1.70×10^{-5}
Barium phosphate	1.3×10^{-29}	Lead(II) iodide	9.8×10^{-9}
Barium sulfate	1.08×10^{-10}	Lead(II) carbonate	7.40×10^{-14}
Calcium carbonate	3.36×10^{-9}	Lead(II) hydroxide	1.43×10^{-15}
Calcium hydroxide	5.02×10^{-6}	Lead(II) phosphate	8.0×10^{-43}
Calcium phosphate	2.07×10^{-29}	Lead(II) sulfate	2.53×10^{-8}
Calcium sulfate	4.93×10^{-5}	Magnesium carbonate	6.82×10^{-6}
Copper(II) carbonate	1.4×10^{-10}	Magnesium hydroxide	5.61×10^{-12}
Copper(II) hydroxide	2.2×10^{-20}	Magnesium phosphate	1.04×10^{-24}
Copper(II) phosphate	1.40×10^{-37}	Silver bromide	5.35×10^{-13}
Iron(II) carbonate	3.13×10^{-11}	Silver chloride	1.77×10^{-10}
Iron(II) hydroxide	4.87×10^{-17}	Silver carbonate	8.46×10^{-12}
Iron(III) hydroxide	2.79×10^{-39}	Silver hydroxide	2.0×10^{-8}
Iron(III) phosphate	9.91×10^{-16}	Silver iodide	8.52×10^{-17}
		Silver phosphate	8.89×10^{-17}
		Silver sulfate	1.20×10^{-5}

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{K}(s)$	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ba}(s)$	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ca}(s)$	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Na}(s)$	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mg}(s)$	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	$\text{Al}(s)$	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Mn}(s)$	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Zn}(s)$	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Fe}(s)$	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Ni}(s)$	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Sn}(s)$	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Pb}(s)$	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(g)$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(aq) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.34 V
$\frac{1}{2}\text{O}_2(g) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Cu}(s)$	0.52 V
$\frac{1}{2}\text{I}_2(s) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(aq) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	$\text{Ag}(s)$	0.80 V
$\frac{1}{2}\text{Br}_2(l) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(g) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(g) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(aq) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(g) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

PERIODIC TABLE OF THE ELEMENTS									
KEY									
Atomic Number									
Symbol									
Standard Atomic Weight									
Name									
79 Au Gold									
197.0									

